

Lee–Yang Theorem

Foundations, Proof and Applications

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Theorem 3.25, item 2

In $d = 1$, there is a unique Gibbs state at each $(\beta, h) \in \mathbb{R}^* \times \mathbb{R}$.

Short-range correlations guarantee uniqueness. Physically, thermodynamic quantities usually exhibit some kind of non-analyticity after a phase transition. The thermodynamic limit that undergoes a singularity in some real parameter often corresponds to that parameter ‘approaching’ the zero of the partition function somewhere after the complex plane extension. Mathematically, the behaviour of $\mathbf{Z}(\beta, h)$.

Holomorphic Functions

Let $D \subset \mathbb{C}$ be a domain. A function $f : D \rightarrow \mathbb{C}$ is said to be **holomorphic** if the derivative

$$f'(z) := \lim_{w \rightarrow z} \frac{f(w) - f(z)}{w - z}$$

exists and is finite for every $z \in D$.

Holomorphic functions are synonymous with **analytic** functions in this presentation, i.e.,

$$f(z) = \sum_{n \geq 0} a_n (z - z_0)^n.$$

Cauchy's Integral Theorem

Let $D \subset \mathbb{C}$ be an open, simply connected domain, and let f be a holomorphic function defined on D . Then, for any closed path $\gamma \subset D$, we have

$$\oint_{\gamma} f(z) dz = 0.$$

Theorem

Let $f(z)$ be a holomorphic function defined on a convex open subset $D \subset \mathbb{C}$. Then, f has an antiderivative, i.e., there exists a holomorphic function $F(z)$ on D such that $F'(z) = f(z)$.

Sketch Proof. By Cauchy's Integral Theorem,

$$\lim_{w \rightarrow z} \frac{F(w) - F(z)}{w - z} = \int_0^1 \lim_{w \rightarrow z} f((1-t)w + tz) dt = \int_0^1 f(z) dt = f(z).$$

Theorem

Every convex region is simply connected.

Sketch Proof. We define a homotopy $H : [0, 1] \times [0, 1] \rightarrow A$ by

$$H(s, t) = sp + (1 - s)\sigma(t), \quad s, t \in [0, 1].$$

Then we verify: (i) **Continuity.** (ii) **Well-definedness.** (iii) **Boundary conditions.** A is path-connected and every loop in A is homotopic, A is simply connected.

Corollary

Let $D \subset \mathbb{C}$ be open and simply connected, and let f be holomorphic on D . Then, f has an antiderivative, i.e., there exists a holomorphic function $F(z)$ on D such that $F'(z) = f(z)$.

Theorem

Let f be a holomorphic function on a simply connected open set $D \subset \mathbb{C}$, which has no zeroes on D . Then, there exists a holomorphic function g on D , called **a branch of the logarithm** of f on D , such that $f = e^g$.

Locally Uniformly Bounded

A family of functions $\mathcal{A}$ defined on $\mathbb{C}$ is said to be **locally uniformly bounded** on a subset $D \subset \mathbb{C}$ if, for every point $z \in D$, there exists a real number M and a neighbourhood $\mathcal{U} \subset D$ of z such that

$$|f(w)| \leq M, \quad \forall w \in \mathcal{U}, \forall f \in \mathcal{A}.$$

Schwarz's Lemma

Let f be a holomorphic function defined on the unit disc $D = \{z \in \mathbb{C} : |z| \leq 1\}$ such that $f(0) = 0$ and $|f(z)| \leq 1$ for all $z \in D$. Then $|f'(0)| \leq 1$.

Sketch Proof. Define

$$g(z) = \begin{cases} \frac{f(z)}{z}, & z \neq 0 \\ f'(0), & z = 0 \end{cases}.$$

Vitali's Convergence Theorem

Let $D \subset \mathbb{C}$ be an open, connected subset and $(f_n)_{n \geq 1}$ a sequence of analytic functions on D that are locally uniformly bounded. Suppose that $(f_n)_{n \geq 1}$ converges pointwise to a function f on a set $S \subset D$, where S has a cluster point in D . Then the sequence $(f_n)_{n \geq 1}$ converges locally uniformly to an analytic function f on D .

Rouché's Theorem

Let C be a simple closed contour in $\mathbb{C}$. Suppose f and g are two functions that are analytic inside and on C , and that

$$|f(z)| > |g(z)|, \quad \forall z \in C.$$

Then f and $f + g$ have the same number of zeroes inside C , counting multiplicities.

Hurwitz Theorem

Let $D \subset \mathbb{C}$ be an open set and suppose that $(f_n)_{n \geq 1}$ is a sequence of analytic functions on D , which converge locally uniformly to an analytic function f on D . Assume that $f_n(z) \neq 0$ for all $z \in D$ and all $n \geq 1$. Then f is either identically zero or $f(z) \neq 0$ for all $z \in D$.

Weierstrass' Theorem on Uniformly Convergent Series of Analytic Functions

Let $D \subset \mathbb{C}$ be a domain. For each $k \in \mathbb{N}$, let $f_k : D \rightarrow \mathbb{C}$ be an analytic function. Suppose that the series $f(z) := \sum_{k=0}^{\infty} f_k(z)$ is uniformly convergent on every compact subset $K \subset D$. Then $f(z)$ is an analytic function on D . Moreover, for each $n \in \mathbb{N}$, the series of derivatives $\sum_{k=0}^{\infty} f_k^{(n)}(z)$ converges uniformly on every compact subset $K \subset D$, and

$$f^{(n)}(z) = \sum_{k=0}^{\infty} f_k^{(n)}(z), \quad \forall z \in D.$$

Theorem 3.6

In the thermodynamic limit, the pressure

$$\psi(\beta, h) := \lim_{\Lambda \uparrow \mathbb{Z}^d} \psi_{\Lambda}^{\#}(\beta, h)$$

is well defined and independent of the sequence $\Lambda \uparrow \mathbb{Z}^d$ and of the type of boundary condition. Moreover, ψ is convex (as a function on $\mathbb{R}^* \times \mathbb{R}$) and is even as a function of h .

Lee–Yang Theorem

Let $\beta \geq 0$. Let $D \subset \mathbb{C}$ be an open, simply connected set such that $D \cap \mathbb{R}$ is an interval of $\mathbb{R}$. Suppose that, for every finite volume $\Lambda \in \mathbb{Z}^d$, the finite-volume partition function with free boundary condition (FBC) satisfies

$$Z_{\Lambda; \beta, h}^{\emptyset} \neq 0, \quad \forall h \in D.$$

Then the (infinite-volume) Ising-model pressure $\psi(\cdot)$ admits an analytic continuation to the domain D .

Proof.

Branch of the logarithm via zero-free condition

$\mathbf{z}_{\Lambda;\beta,h}^\emptyset \neq 0$. Show the existence of a branch of the logarithm on a simply connected, zero-free domain. Thus,

$$f(h) = \mathbf{z}_{\Lambda;\beta,h}^\emptyset \implies f(h) \neq 0 \text{ on } D,$$

so there exists an analytic function $\log \mathbf{z}_{\Lambda;\beta,h}^\emptyset$ defined on the whole of D .

Infinite-volume limit and g_n

Define

$$g_n(h) := \exp\left(\frac{1}{|\Lambda_n|} \log \mathbf{Z}_{\Lambda_n; \beta, h}^{\emptyset}\right) = \exp(\psi_{\Lambda_n}^{\emptyset}(h)),$$

where $\psi_{\Lambda_n}^{\emptyset}(h)$ is the finite-volume pressure with FBC. By **Theorem 3.6**, the infinite-volume pressure

$$\psi(\beta, h) = \lim_{\Lambda \uparrow \mathbb{Z}^d} \psi_{\Lambda}^{\#}(\beta, h)$$

exists for all real $h \neq 0$ and is independent of both FBC and the sequence $\Lambda \uparrow \mathbb{Z}^d$, and

$$g_n(h) = \exp(\psi_{\Lambda_n}^{\emptyset}(h)) \xrightarrow{n \rightarrow \infty} \exp(\psi(\beta, h)) =: g(h).$$

Locally uniformly bounded

The finite-volume Hamiltonian with FBC is

$$\mathbf{z}_{\Lambda_n; \beta, h}^{\emptyset} = \sum_{\omega \in \Omega_{\Lambda_n}} \exp\left(-\mathcal{H}_{\Lambda_n; \beta, h}^{\emptyset}(\omega)\right).$$

For $h \in D$, we have

$$\left| \mathbf{z}_{\Lambda_n; \beta, h}^{\emptyset} \right| \leq \exp((2d\beta + |\Re h| + \log 2)|\Lambda_n|).$$

Dividing by $|\Lambda_n|$ in the exponent and then exponentiating again for $g_n(h)$, we obtain

$$|g_n(h)| = \exp\left(|\Lambda_n|^{-1} \log |\mathbf{z}_{\Lambda_n; \beta, h}^{\emptyset}|\right) \leq \exp(2d\beta + |\Re h| + \log 2).$$

Since this bound depends only on the real part of h (and not on n), $\{g_n\}$ is locally uniformly bounded in D .

Applying theorems

- By **Vitali's convergence theorem**, $\{g_n\}$ converges locally uniformly on the entire domain D to some analytic limit g .
- By **Hurwitz theorem**, the limit g also has no zeroes in D .
- By the theorem of branch of the logarithm to conclude that there is an analytic logarithm $\log g(h) = \log(e^{\psi(h)}) = \psi(h)$ on D .

Therefore, we define $\psi(h) := \log g(h)$ for all $h \in D$, which provides the analytic continuation of the Ising model pressure from real $h \in D \cap \mathbb{R}$ to the entire simply connected set D .

Lee–Yang Circle Theorem

Let $\beta \geq 0$. Consider the open half-planes

$$H^+ = \{h \in \mathbb{C} : \Re h > 0\}, \quad H^- = \{h \in \mathbb{C} : \Re h < 0\}.$$

Then, for any finite volume $\Lambda \in \mathbb{Z}^d$ and FBC $\emptyset$, the finite-volume partition function $\mathbf{Z}_{\Lambda; \beta, h}^\emptyset$ does not vanish anywhere in H^+ or H^- . Then, the pressure admits an analytic continuation.

After reparametrising by $z = e^{-2h}$, all zeroes of the partition function lie on the unit circle $\{z \in \mathbb{C} : |z| = 1\}$ rather than in the open unit disc $\mathcal{U} = \{z : |z| < 1\}$.

Ising Model

Spins $\sigma_x \in \{-1, 1\}$ on each site x of a finite lattice $\Lambda \subset \mathbb{Z}^d$. The pairwise interaction between these spins is described by the Hamiltonian

$$\mathcal{H}_\Lambda(\sigma_\Lambda) = - \sum_{\{x,y\} \in \Lambda} J_{xy} \sigma_x \sigma_y - \sum_{x \in \Lambda} h \sigma_x,$$

where the couplings J_{xy} are pairwise ferromagnetic (i.e., non-negative) interactions, and $h \in \mathbb{C}$ is an external magnetic field with $\Re h \neq 0$. The partition function is given by

$$\mathbf{Z}_{\Lambda, \beta}(h) = \sum_{\sigma \in \{-1, 1\}^\Lambda} \exp(-\beta \mathcal{H}_\Lambda(\sigma_\Lambda)).$$

Theorem

If the ferromagnetic couplings satisfy

$$J_{xy} = J_{yx} = J_{x-y,0} \geq 0, \quad \forall x, y,$$

and the *a priori* measure μ_0 on $\mathbb{R}$ satisfies:

- (i) μ_0 is even, i.e. $\mu_0(d\sigma)$ is invariant under $\sigma \mapsto -\sigma$.
- (ii) Its Laplace transform

$$\hat{\mu}_0(h) = \int_{\mathbb{R}} e^{h\sigma} d\mu_0(\sigma) \neq 0, \quad \forall h \in \mathbb{H}_+ := \{z \in \mathbb{C} : \Re z \neq 0\}.$$

Then, for any choice of external fields $\{h_x\}_{x \in \Lambda}$ with $\Re h_x > 0$ for all $x \in \Lambda$, the partition function satisfies

$$\mathbf{Z}_{\Lambda}(\{h_x\}) := \int_{\mathbb{R}^{|\Lambda|}} \exp(-\mathcal{H}_{\Lambda}(\sigma_{\Lambda})) \prod_{x \in \Lambda} d\mu_0(\sigma_x) \neq 0.$$

Connected Correlation Functions

For any finite region $\Lambda \subset \mathbb{Z}^d$, inverse temperature $\beta > 0$, and homogeneous magnetic field h , the **connected correlation** of n spins is defined by

$$\langle \sigma_{x_1}; \dots; \sigma_{x_n} \rangle_{\Lambda; \beta, h}^c = \left[\frac{\partial^n}{\partial \varepsilon_1 \cdots \partial \varepsilon_n} \log \left(\left\langle \exp \left[\sum_{i=1}^n \varepsilon_i \sigma_{x_i} \right] \right\rangle_{\Lambda; \beta, h} \right) \right]_{\varepsilon=0},$$

where $\langle \cdot \rangle_{\Lambda; \beta, h}$ denotes the thermal average with respect to the (finite-volume) Gibbs measure associated with $\mathcal{H}_\Lambda$, and the superscript c denotes the connected correlation function.

Remark

The connected correlation functions $\langle \sigma_{x_1}; \dots; \sigma_{x_n} \rangle^c$ can be viewed as the higher-order cumulants of the random variables $\sigma_{x_1}, \dots, \sigma_{x_n}$. Formally, they arise from taking derivatives of the log of the moment generating function, thereby removing contributions from lower-order correlations. In particular, for $n = 2$, we recover the usual covariance-like form $\langle \sigma_x; \sigma_y \rangle^c = \langle \sigma_x \sigma_y \rangle - \langle \sigma_x \rangle \langle \sigma_y \rangle$.

Main Theorem

Analyticity of Connected Correlations

Under the assumptions above for the Ising Hamiltonian $\mathcal{H}_\Lambda$, with homogeneous magnetic field h and ferromagnetic couplings satisfying

$$J_{xy} = J_{yx} = J_{x-y,0} \geq 0, \quad \forall x \neq y, \quad \sum_{x \neq 0} J_{0x} < \infty.$$

Let $\langle \sigma_{x_1}; \dots; \sigma_{x_n} \rangle_{\Lambda; \beta, h}^c$ be the connected correlation function of order $n \in \mathbb{N}$. Then, for any collection of sites $x_1, \dots, x_n \in \mathbb{Z}^d$ and for every $\beta > 0$, the thermodynamic limit

$$\langle \sigma_{x_1}; \dots; \sigma_{x_n} \rangle_{\beta, h}^c := \lim_{\Lambda \nearrow \mathbb{Z}^d} \langle \sigma_{x_1}; \dots; \sigma_{x_n} \rangle_{\Lambda; \beta, h}^c$$

exists and defines a function of the external field h that is analytic in the half-planes $\mathbb{H}_+$.

Sketch Proof.

- Setting up a modified partition function. When $\Re(h_x) > 0$,

$$\mathbf{Z}_\Lambda \left(h + \sum_{\alpha=1}^n \varepsilon_\alpha e^{ik_\alpha x} \right) \neq 0.$$

$$\mathcal{R}_\Lambda(h, \varepsilon) := \mathbf{Z}_\Lambda(h, \varepsilon)^{\frac{1}{|\Lambda|}} := \exp[f_\Lambda(h, \varepsilon)].$$

- Uniform bounds and thermodynamic limit.

$$\sup_{\Lambda; (h, \varepsilon) \in K} |\mathcal{R}_\Lambda(h, \varepsilon)| < \infty.$$

By **Vitali's convergence theorem**, $\mathcal{R}_\Lambda \rightarrow \mathcal{R}_\infty$ uniformly on every compact $K \subset D$. By **Weierstrass' theorem**, $\mathcal{R}_\infty$ is analytic on D .

- Non-vanishing of the limit. Let $g_\lambda^1(z) = \mathcal{R}_\lambda(h, z, 0, \dots, 0)$ converges to $g_\infty^1(z) = \mathcal{R}_\infty(h, z, 0, \dots, 0)$ uniformly, by **Hurwitz theorem**, $\mathcal{R}_\infty(h, \epsilon) \neq 0$ for $\Re \epsilon h > \sum_\alpha |\epsilon_\alpha|$.

$$f_\infty(h, \epsilon) := \log \mathcal{R}(h, \epsilon).$$

- Connection to correlation functions.

$$\langle \sigma_{x_1}; \dots; \sigma_{x_n} \rangle_{\beta, h}^c := \lim_{\Lambda \nearrow \mathbb{Z}^d} \langle \sigma_{x_1}; \dots; \sigma_{x_n} \rangle_{\Lambda; \beta, h}^c$$

is an analytic function of h on $\{\Re \epsilon h \neq 0\}$.

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